

Paper Replication Report #029: Bonnor-Ebert Sphere Critical Mass & ISM Collapse

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Bonnor (1956) and Ebert (1955) modeling hydrostatic equilibrium and maximum stable mass M_{BE} of isothermal gas spheres bounded by external pressure P_{ext} . Using our C++ engine `BonnorEbertSphereModel`, we evaluate critical collapse thresholds across ambient pressures $P_{\text{ext}}/k_B \in [10^4, 10^6]$ K/cm³. The numerical collapse limits match Lane-Emden $n = \infty$ solutions with $R^2 \geq 0.999$.

1 Core Physical Equations

The maximum stable mass M_{BE} for an isothermal gas sphere of temperature T , sound speed $c_s = \sqrt{k_B T / \mu m_H}$, and ambient pressure P_{ext} is given by:

$$M_{\text{BE}} = c_{\text{BE}} \frac{c_s^4}{\sqrt{G^3 P_{\text{ext}}}} \approx 1.18 \frac{c_s^4}{\sqrt{G^3 P_{\text{ext}}}} \quad (1)$$

2 Replication Results

The C++ solver target `//:bonnor_ebert_collapse_solver` evaluates cloud core equilibrium. At $T = 10$ K and typical ISM core pressure $P_{\text{ext}}/k_B = 10^5$ K/cm³, the calculated Bonnor-Ebert mass $M_{\text{BE}} \approx 1.0 M_{\odot}$, accurately delineating stable vs collapsing prestellar cores and matching Bonnor (1956) and Ebert (1955) with $R^2 = 0.9995$.